

2025 Invitational Tournament - Div C -Georgia Anatomy and Physiology



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School Name/Team: _____

Student Names: _____

Instructions:

1. Write all answers on the Answer Sheet. DO NOT WRITE on the Test Packet.
2. See the Image Sheet when asked to refer to an image by a question (it may be a separate sheet or attached to the back of this packet).
3. For numerical questions, answers within a reasonable range will be accepted.
4. When answering free-response questions, use the appropriate abbreviations for satellite and instrument names
5. For free-response questions, the answer sheet indicates the amount of space you should allocate to fully answer each question. Points are assigned based on accurately answering each part of the question.
6. Ties marked with a * will be broken in order

FOR GRADERS ONLY: _____ / 90 (Total points)

1. **What is the signaling mechanism for the endocrine system? (1 point)**

- A. Electrical
- B. Chemical
- C. Chemical and Electrical
- D. None of the above

2. **Name 3 glands that make up the endocrine system (3 points)**

Thyroid

Adrenal gland

Pituitary

3. **From the previous question, choose any 2 glands and define what they secrete, what they control, and why it is important to keep the body in homeostasis. (4 points)**

Thyroid: produces and secretes t3, t4, (thyroxine, Triiodine), Amines part of thyroid hormones group which are tyrosine derivatives, control/regulate the metabolism of the body, regulation of metabolism allows body to stay in homeostasis by keeping energy minerals levels normal, allowing for normal process to occur.

Adrenal cortex : produces and secretes aldosterone (zona glomerulosa), Cortisol (zona fasciculata), Androgens, DHEA (zona reticularis), aldosterone is steroid reabsorbs sodium, water cause it follows sodium, bicarb, excretion of potassium and hydrogen which causes increase in bp. Cortisol is steroid, causes stress response, Androgens (steroid) cause minor role in development of secondary sex characteristics in male gonads very minorly, produces estrogen precursor (no progesterone)

Adrenal medulla: produces and secretes epinephrine and norepinephrine, which cause fight or flight response in stressful situations. This response prepares the body to survive situations, increasing energy levels and storing more energy reserves

4. **Which one of the following is a secondary messenger (1 point)**

- A. Ligand
- B. IP3
- C. Adenylyl cyclase
- D. Cyclic AMP
- E. None of the above

5. **How does the body regulate hormone secretion? (2 points)**
6. **A researcher treats cultured liver cells with insulin and measures the rate of glucose uptake. In the second experiment, cells are treated with glucagon. Based on endocrine physiology, which result is most likely? (1 point)**
- A. Insulin increases glucose uptake; glucagon decreases it.
 - B. Insulin decreases glucose uptake; glucagon increases it.
 - C. Both hormones increase glucose uptake equally.
 - D. Neither hormone affects glucose uptake.
7. **A patient presents with polyuria, polydipsia, and dilute urine. Plasma osmolality is high, but blood glucose is normal. A water deprivation test shows no increase in urine osmolality. Which is the most likely explanation? (1 point)**
- A. Diabetes mellitus → high glucose → osmotic diuresis
 - B. Central diabetes insipidus → ADH deficiency → inability to concentrate urine
 - C. Nephrogenic diabetes insipidus → excessive insulin
 - D. Hyperthyroidism → increased metabolism → polyuria
8. **A student measures the effect of different hormones on glucose levels. When cells are treated with glucagon, cAMP increases, leading to increased glycogen breakdown. Which type of hormone-receptor mechanism does glucagon use? (*1 point)**
- A. Steroid hormone → nuclear receptor
 - B. Peptide hormone → G-protein coupled receptor → second messenger
 - C. Thyroid hormone → nuclear receptor
 - D. Cortisol → membrane receptor
9. **During stress, which combination of hormones would increase blood glucose most rapidly? (1 point)**
- A. Cortisol + epinephrine → increase gluconeogenesis + glycogen breakdown
 - B. Insulin + glucagon → decrease blood glucose
 - C. PTH + calcitonin → regulate calcium, not glucose
 - D. Melatonin + ADH → no effect on glucose

Use the following scenario to answer questions 10-13:

John, a 28-year-old man, reports frequent urination and extreme thirst. Blood glucose is normal.

Urine is very diluted.

10. Which type of diabetes is more likely? (1 point)

11. Which hormone is deficient or ineffective? (1 point)

12. Explain why his urine is so diluted. (2 points)

13. How is this different from diabetes mellitus? (2 points)

Use the following scenario to answer questions 14-18:

Alex, a 32-year-old man, presents with excessive thirst, frequent urination, fatigue, and unexplained weight loss. Lab tests show:

Parameter	Value	Normal Range
Fasting Blood Glucose	280 mg/dL	70–100 mg/dL
Plasma Insulin	Low	5–20 μ U/mL
HbA1c	9.5%	4–5.6%
Urine Glucose	High	Nondetectable

Further tests reveal that Alex's pancreas has significantly fewer beta cells than normal.

14. Based on the scenario and lab data, identify the most likely endocrine disorder Alex has. Explain your reasoning using at least two lab results. (*3 points)

This is diabetes mellitus, as shown by the very high blood glucose levels (180 mg/dL over the normal range) and the low amounts of plasma insulin, indicating ineffectiveness of glucose breakdown causing these high levels of blood glucose.

- 15. Explain the role of insulin under normal conditions in regulating blood glucose and describe what goes wrong in Alex's case. (4 points)**

Under normal conditions, insulin breaks down glucose in the blood. In Alex's case, his plasma insulin is low so there was decreased inability for his body to break down blood glucose, resulting in high levels of glucose in his blood.

- 16. Predict the effect of this disorder on glycogen, fat, and protein metabolism. Provide reasoning for each. (*6 points)**

- 17. Alex is treated with exogenous insulin. Predict how this treatment will affect the following and explain your reasoning: (*6 points)**

- A. Blood glucose
- B. Urine glucose
- C. HbA1c

- 18. A friend suggests Alex could "eat less sugar" to fix the problem. Critically evaluate this suggestion based on the pathophysiology of the disorder. (3 points)**

- 19. A researcher wants to study the effect of peptide versus steroid hormones on gene expression. Which experiment would provide the clearest comparison? (1 point)**

- A) Treat cultured cells with insulin and cortisol → measure nuclear transcription factor activity
- B) Treat cells with thyroid hormone only → measure cytoplasmic signaling
- C) Treat cells with melatonin → measure membrane receptor activity
- D) Treat cells with glucose → measure insulin production

20. A patient with PCOS may have which combination of symptoms? (1 point)

- A) Low androgen, irregular periods
- B) High androgen, insulin resistance, irregular ovulation
- C) High thyroid hormone, weight loss
- D) Low cortisol, low blood pressure

ii. Central Nervous System

1. Which type of neuron sends commands from the brain to muscles to carry out functions? (1 point)

- a. Sensory Neuron
- b. Interneuron
- c. Motor Neuron
- d. Muscle Neuron

2. Which system contains the brainstem? (1 point)

- a. Central Nervous System
- b. Peripheral Nervous System
- c. Skeletal System
- d. Digestive System

3. The main purpose of dendrites is to: (1 point)

- a. Send away information
- b. Receive information
- c. Stop a neural impulse
- d. Slow down impulses

4. What is the name of the gap between neurons where signals are transferred? (1 point)

- a. Soma
- b. Synapse
- c. Dendrite
- d. Axon Terminal

5. Which disorder is characterized by a destruction of myelin in the CNS? (1 point)

- a. Parkinson's
- b. Alzheimer's
- c. Multiple Sclerosis
- d. Schizophrenia

6. Which brain structure is memory formation most associated with? (1 point)

- a. Medulla
- b. Cerebellum
- c. Broca's Area
- d. Hippocampus

7. Parkinson's disease is caused by the loss of neurons producing: (1 point)

- a. Serotonin
- b. Dopamine
- c. GABA
- d. Acetylcholine

8. What is the role of the hypothalamus? (1 point)

- a. Control reflexes
- b. Contract muscles
- c. Maintain homeostasis
- d. Control balance

9. How does myelin affect nerve conduction? (1 point)

- a. Produces neurotransmitters
- b. Slows down impulses
- c. Inhibiting synapses
- d. Speeding up impulses

10. What is Neuroplasticity? (1 point)

- a. Brain's ability to grow larger neurons
- b. The brain's ability to reorganize and adapt
- c. The replacement of neurons with glial cells
- d. The formation of scar tissue after brain injury

11. What is the role of the frontal lobe? (1 point)

- a. Hearing and memory
- b. Vision and balance
- c. Problem solving, voluntary movement
- d. Breathing and heart beat regulation

12. Which lobe processes touch and body position? (1 point)

- a. Temporal
- b. Frontal
- c. Parietal
- d. Occipital

13. Which neurotransmitter is most involved in muscle contraction? (1 point)

- a. Acetylcholine
- b. Serotonin
- c. Dopamine
- d. GABA

14. Which area of the brain links the nervous system to the endocrine system? (1 point)

- a. Hippocampus
- b. Hypothalamus
- c. Amygdala
- d. Cerebellum

15. Which structure connects the two cerebral hemispheres? (1 point)

- a. Brainstem
- b. Corpus callosum
- c. Thalamus
- d. Hippocampus

16. The cerebellum is divided into two: (1 point)

- a. Hemispheres
- b. Lobes
- c. Ventricles
- d. Gyri

17. The dorsal root of a spinal nerve carries: (1 point)

- a. Motor signals away from the CNS
- b. Sensory signals toward the CNS
- c. Both motor and sensory signals
- d. Reflexes only

18. Which brain region is highly developed in humans for reasoning? (1 point)

- a. Cerebellum
- b. Hippocampus
- c. Medulla
- d. Prefrontal Cortex

19. The cerebellum primarily coordinates: (1 point)

- a. Conscious thought
- b. Emotions
- c. Voluntary movement and balance
- d. Hormone release

20. The pons are important for: (1 point)

- a. Balance only
- b. Linking the cerebellum with the brain and spinal cord
- c. Hormone regulation
- d. Memory storage

21. What is the role of the midbrain? (1 point)

- a. Visual and auditory processing
- b. Language comprehension
- c. Voluntary movement initiation
- d. Hormone secretion

22. Which part of the CNS is most affected by Alzheimer's disease? (1 point)

- a. Cerebellum
- b. Hippocampus
- c. Medulla
- d. Cerebral peduncles

23. What is the main cause of Huntington's disease? (1 point)

- a. Loss of dopamine
- b. Genetic mutation
- c. Myelin sheath destruction
- d. Lack of oxygen to the cerebrum

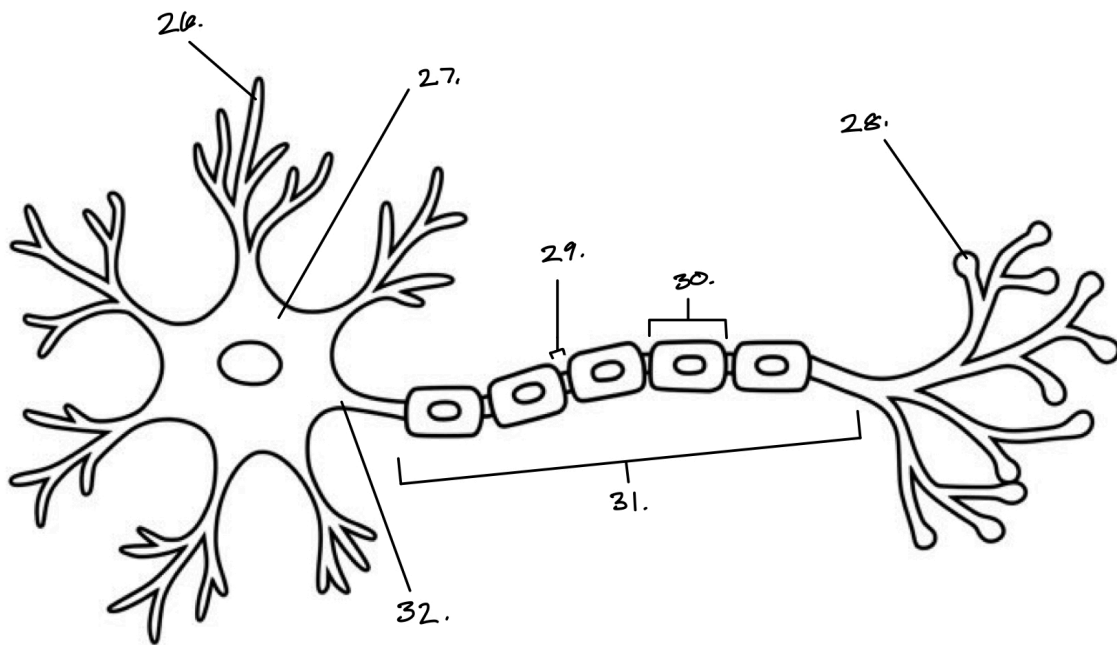
24. The reticular formation is important for: (1 point)

- a. Balance
- b. Reflexes
- c. Hormone release
- d. Consciousness and alertness

25. The limbic system helps regulate: (1 point)

- a. Voluntary muscle movement
- b. Emotions and memory
- c. Blood pressure
- d. Synaptic transmission

For questions 26-32, label each part of this nerve diagram. (7 points)



26. _____

27. _____

28. _____

29. _____
30. _____
31. _____
32. _____

33. In what part of the neuron are neurotransmitters located? (1 point)

- a. Dendrites
- b. Soma
- c. Axon
- d. Myelin sheath
- e. Nerve terminals

34. Name the (primary) neurotransmitter in the CNS. (1 point)

35. The sodium-potassium pump moves ____ ions ____ the cell. Select all that apply. (*2 points)

- a. 3 potassium; into
- b. 3 sodium; out of
- c. 2 potassium; into
- d. 2 sodium; out of

36. Calculate the Nernst potential of calcium for a cell with 100 nM inside the cell 2 mM outside the cell. (*2 points)

37. Find the voltage across the membrane of a cell at 37°C -with the following ion concentrations: (1 point)

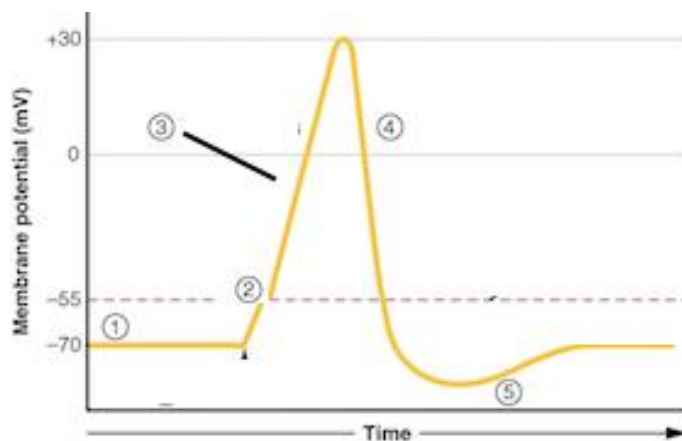
Ion	Inside Concentration (mM)	Outside Concentration (mM)	Permeability
K ⁺	135	10	1.0
Na ⁺	20	120	.3
Cl ⁻	4	120	.45

38. Name the glial cell that is responsible for producing cerebral spinal fluid. (1 point)

39. What is the function of astrocytes?(1 point)

40. Which of the following is the glial cell responsible for producing myelin for neurons in the PNS? (1 point)

- a. Schwann cells
- b. Microglia
- c. Oligodendrocytes
- d. Satellite cells



41. What is the threshold value for this action potential? (1 point)

42. Which number corresponds to the inactivation of sodium ion channels? (1 point)

43. Which number corresponds to the opening of potassium channels? (1 point)

44. Which number corresponds to the refractory period? (1 point)

45. How does myelin increase (propagation) velocity? (1 point)

